# Diverse Style Prompt Assistant for Generalized Remote Sensing Object Classification

Diedie Jia, Wenda Zhao, *Member, IEEE*, Hao Liu, and Haipeng Wang

***Abstract*—Due to the existence of domain shift between different remote sensing datasets, remote sensing object classification models trained on the source domain often face the challenge of reduced accuracy when applied to unseen domains. In this article, we propose a diverse style prompt assistant for generalized remote sensing object classification. We utilize the powerful knowledge reserve of the CLIP model to solve the domain generalization (DG) problem from the perspective of text prompts. CLIP's text features simulate feature-space style diversity, and then diverse styles guide the classification model to generate general image features. Specifically, we first implement a style diversity loss and a maximum entropy loss to train prompts to obtain multiple style prompts. Then, we introduce the style prompts to guide the optimization process of image features cross-modally within the joint feature space, enabling the classification model to generate domain-invariant features. We conduct extensive experiments on three remote sensing datasets, and the results show that our method achieves superior generalization performance compared with the state of the art. Codes are available at https://github.com/Jia-dd/DSPA**

***Index Terms*—Diverse text prompt generation, domain generalization (DG), domain-invariant features, remote sensing object classification.**

## I. Introduction

REMOTE sensing object classification is a key step in acquiring remote sensing image information, which can provide auxiliary information for many downstream tasks. In recent years, research on remote sensing object classification has become increasingly in-depth [1], [2]. However, due to the particularity of remote sensing image acquisition, different device parameters, lighting conditions, and radiation environments during the image acquisition process can lead to significant differences in color distribution, texture, lighting, and contrast of the collected images, which are also known as style differences. The difference in style between different datasets is called the domain gap, which leads to a significant decrease in the classification accuracy of models trained on source domains when applied to other unseen domains. The domain generalization (DG) task is proposed to address the issue of decreased accuracy caused by the domain gap. It aims to improve the generalization of the model so that it can maintain high accuracy when applied to unseen domains with different styles from the source domain.

Due to the existence of domain differences between different domains and the inability of the model to obtain any information from data other than the source domain during training, DG tasks still pose significant challenges. The accuracy of the model trained on the training domain will decrease significantly when applied to unseen domains, thereby limiting the scope of the model's application. Therefore, improving the generalization of classification models is of great significance for their widespread application.

There have been many studies in the field of DG at present. According to different perspectives on problem-solving, existing methods can be divided into three categories: representation learning [3], [4], data manipulation [5], [6], and learning strategy [7], [8], [9]. Representation learning is currently the most popular method in the field of DG, which mainly includes two processing methods: domain-invariant representation learning and feature disentanglement. Data manipulation focuses on manipulating input data through methods such as data augmentation or data generation to assist models in learning general representations. Learning strategy-based methods mainly promote the generalization ability of models by utilizing various learning strategies, including but not limited to ensemble learning, meta-learning, gradient operations, and self-supervised learning, etc.

In recent years, large language models have gradually attracted more and more attention and have been applied in various fields by combining with other technologies. In terms of image processing, large language models can leverage the richness and accessibility of text to assist visual models in learning general representations. Among them, CLIP [10] is the first to combine large language models with image classification tasks. CLIP utilizes a large number of image–text pairs as training data to fully train the text encoder and image encoder, and has strong feature extraction capabilities that can be flexibly transferred to other downstream tasks. However, due to the particularity of remote sensing images, CLIP does not perform well on remote sensing image datasets. Moreover, the CLIP model has a huge number of parameters, so directly fine-tuning CLIP would consume a significant amount of time and computational resources, which is clearly impractical. In addition, due to the existence of a domain gap between remote sensing image datasets, even if CLIP is fine-tuned directly using the source domain without considering time and computational limitations, the performance of the fine-tuned

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Diedie Jia and Wenda Zhao are with the School of Information and Communication Engineering, Dalian University of Technology, Dalian 116024, China (e-mail: jia1234@mail.dlut.edu.cn; zhaowenda@dlut.edu.cn).

Hao Liu is with the School of Integrated Circuits, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: liuhao2020@sjtu.edu.cn).

Haipeng Wang is with the Unit 92728 of PLA, Shanghai 200436, China (e-mail: whp5691@163.com).

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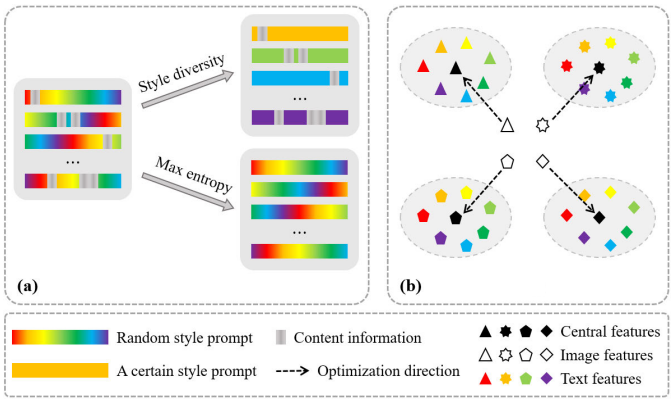



Fig. 1. Overall framework of our work. The proposed method is divided into two stages: (a) style prompt generation and (b) classification model training. We first generate diverse style prompts using multiple loss constraints in the first stage. Then, in the second stage, we use the obtained style prompts to generate diverse text features and provide constraints for the optimization process of the classification model.

model will still significantly decrease when applied to unseen domains.

In order to improve the DG capability of CLIP, PromptStyler [11] synthesizes different styles by changing the prompts to simulate various distribution shifts in the joint space, and trains a classifier using the synthesized text features with different styles to achieve model transfer. RISE [12] distills CLIP's knowledge into the student network through knowledge distillation, to provide guidance for the regularization of the student model during training. However, PromptStyler only trains a simple classification head, and the knowledge that can be learned is very limited. The text prompts used by RISE are fixed text templates, which have limitations in describing image styles.

Although it is difficult to obtain images with diverse styles for the DG task, it is easy to obtain text prompts representing various styles by changing the text description, thus generating diverse text features in the feature space. The image features and text features generated by CLIP are aligned in the joint feature space. If diverse text features can be obtained, it is equivalent to obtaining diverse image features. Therefore, as long as the image features generated by the classification model are aligned with these text features, the generalization of the classification model can be achieved without using a large number of source domains.

Based on the above analysis, we propose a method for learning style prompts through entropy maximization to simulate the diverse distribution of the joint feature space and achieve DG. As shown in Fig. 1, our method mainly consists of two stages. In the first stage, we first use learnable style word vectors to replace fixed style words. Then, we add style diversity loss to the text features and perform entropy maximum constraint on the outputs of the classification head to generate prompts with multiple different styles. In the second stage, the generated text prompts are used as inputs to the text encoder to obtain diverse text features in the joint feature space, and they are used as auxiliary to train the classification model. Specifically, we use text features to obtain the central feature, which is the general representation of text features. We compare the image features generated by the classification model with the central features to obtain the corresponding logits and calculate the cross-entropy loss between the logits and the labels. In addition, in order to make the learning process of the classification model converge faster, we use the image features of CLIP to perform distillation learning on the classification model at the feature level and logit level, respectively, to ensure that the classification model can better learn the feature extraction ability of CLIP. By utilizing the richness of textual information, the generalization of the classification model can be improved with only a single source domain.

The main contributions of this work can be summarized as follows.

1) We design a style diversity loss to constrain the diversity of prompts, and propose a maximum entropy loss to ensure that only style information is included in the prompts, ultimately obtaining diverse style prompts to simulate feature diversity in the feature space.

2) We utilize text features that contain different styles and content information to guide the classification model to generate general image features in the joint feature space, improving the generalization of the classification model.

3) Abundant experiments are conducted on three widely used remote sensing datasets, and the results validate the effectiveness of the proposed method.

## II. Related Work

### A. Domain Generalization

The differences in image styles between different datasets result in a domain gap, which reduces the accuracy of the model when transferred between different datasets. To address this issue, Wang et al. [13] design an unsupervised adaptive classification network that mitigates the negative impact of domain gap through a multikernel domain aligner. Feng et al. [14] implement a multimodal deep adaptive classification network using a middle fusion strategy, effectively solving the problem caused by the domain gap. MUCCM [15] improves the cross-domain stability of the model by integrating knowledge from multiple source domains. However, in some more extreme scenarios, we may not be able to obtain any information about the target domain. To overcome this, the concept of DG was introduced, aiming to learn from source domains to maintain high accuracy on unseen domains with a distribution gap. Based on the number of source domains used in training, DG can be classified into two types: multisource and single-source DG. Multisource DG [16], [17], [18] can use multiple related but distinct source domains during training. In addition to the label of the images, each source domain has a domain label. By learning general representations that are invariant to different marginal distributions, the model's generalization ability can be improved. Single-source DG [19], [20] assumes that the model cannot obtain domain labels during training, and all samples are sampled from the same domain, which makes the task of single-source DG more challenging.

There are currently three mainstream approaches for solving DG problems: representation learning, data manipulation, and

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learning strategies. Representation learning aims to enable models to obtain better feature extraction capabilities through domain-invariant representation learning and feature disentanglement. For example, Fan et al. [3] improve the model's generalization ability by learning to adapt to the data from different domains. Zhao et al. [7] design a style-content metric learning framework to guide the model to use image content information rather than style information. Data manipulation methods increase the diversity of existing training samples. FSDR [5] randomizes images in frequency space, keeping domain-invariant frequency components fixed and changing domain-variant frequency components. Chen et al. [6] propose a center-aware adversarial enhancement technique, which expands the source domain distribution by changing the samples of the source domain. The learning strategy-based approach focuses on utilizing various learning strategies to improve the generalization ability of the model. Zhao et al. [8] design a memory-based multisource meta-learning framework to simulate the train-test process in DG tasks. Kim et al. [9] propose a new regularization method, which utilizes a class-specific domain perturbation layer to complete contrastive learning using only positive data pairs.

In this work, we start from the perspective of representation learning and use the diverse text features generated by CLIP to constrain the learning process of the classification model in the joint feature space. We use feature constraints in the joint feature space to enable the feature encoder to generate more general domain-invariant representations, reducing the impact of the domain gap on the classification model.

### B. Multimodal Learning

Most visual task heavily relies on labeled image data and require task-specific pretraining or fine-tuning. In recent years, vision-language models have been increasingly applied [21]. By utilizing large-scale image–text pairs for pretraining, visual-language models can learn to generate general visual representations that can be transferred zero-shot to many downstream tasks. Most visual-language models typically use contrastive loss to align the image features and the text features in a joint feature space [22], [23]. They learn perception from a large number of publicly available image–text pairs, thus possessing strong feature extraction capabilities. At present, visual-language models have been applied in various fields, such as image classification [24], object detection [25], [26], semantic segmentation [27], [28], and point cloud classification [29].

CLIP is a typical visual-language model trained on 400 million image–text pairs by predicting the correct pairing of images and text. It demonstrates strong zero-shot transfer capability, achieving ResNet-101-level performance on ImageNet without fine-tuning. There are many other visual-language models besides CLIP. Yang et al. [22] propose triple contrastive learning, which utilizes cross-modal and intramodal self-supervision for visual pretraining. Lu et al. [30] conduct prompt distribution learning to assist pretrained visual-language models in adapting to downstream tasks. Zhang et al. [31] propose a training-free adaptation method, tip-adapter, which utilizes a key-value cache model from the few-shot training set to construct an adapter, and updates the prior knowledge encoded in CLIP through feature retrieval. Dunlap et al. [32] learn the transformation of the image embeddings from the training domain to each unseen test domain while preserving task-relevant information.

Based on the inspiration from existing work, we utilize the alignment of image and text features of visual-language models, and generate diverse text features by changing text prompts to simulate the diversity of the joint feature space. We utilize the diversity of feature spaces to guide the classification model to learn general image features, thereby improving its generalization ability.

### C. Knowledge Distillation

The concept of knowledge distillation was first proposed by Hinton et al. [33], with the main purpose of enabling a lightweight student model to learn the knowledge possessed by one or more larger teacher models. The knowledge distillation technique optimizes the student model to match the output or intermediate features of the teacher model, thereby transferring knowledge from the teacher to the student model.

Overall, knowledge distillation technology consists of three key components: teacher–student architecture, distillation knowledge, and distillation strategy. There have been various studies on knowledge distillation related to different research focuses. Part of the work is dedicated to researching different teacher–student architectures [34], [35]. Among them, Xie et al. [36] employ a trained EfficientNet as a teacher to train a larger EfficientNet student, then iteratively replace the teacher with the student to enhance the model's generalization capability. Mirzadeh et al. [37] introduce multistep knowledge distillation, using an intermediate-sized network as a teacher assistant to bridge the gap between the student and the teacher. Part of the work studies the diversity of distillation knowledge. For example, Passban et al. [38] consider the importance of each layer in the teacher model, and perform combinatorial distillation on each layer. Chen et al. [39] propose a semantic calibration method, which automatically assigns appropriate target layers of the teacher model to each student layer through an attention mechanism. Another part of the work is to improve distillation efficiency by changing the distillation strategy [40]. Among them, Shen et al. [41] allow the student model to become its own teacher model to achieve self-distillation of the model. Guo et al. [42] propose an efficient online knowledge distillation framework that treats all networks as collaborative students, enabling mutual knowledge transfer among any student model. In addition, there are some works that utilize multiple models as teachers. For example, Xiang et al. [35] propose a novel self-paced knowledge distillation framework to adaptively transfer knowledge from multiple teacher models to the student model.

Our proposed method also employs knowledge distillation. We distill knowledge between the CLIP image encoder's features and the classification model's features, aligning the latter closer to CLIP's joint feature space. In addition, we also apply distillation constraints between the logits generated by image–text feature comparison and the classification logits, thereby enhancing the classification head's parameter learning.

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## III. Method

The biggest challenge for the DG task is the inability to obtain sufficient training data, and the scarcity of training samples greatly limits the performance improvement of the model. Especially for single-source DG, the samples used by the model during training are all from the same domain, lacking diverse information as supervision, resulting in a decrease in the model's generalization performance. With the proposal of the CLIP model, its powerful zero-shot transfer ability has attracted a lot of attention. CLIP achieves strong feature extraction capability by training on a large number of image–text pairs, and the image features and text features generated by CLIP are aligned in the joint feature space. However, due to the particularity of remote sensing images, although CLIP can easily adapt to many downstream tasks, it performs poorly in the field of remote sensing object classification. In addition, due to the large number of parameters in the CLIP model, directly using remote sensing images to fine-tune CLIP not only consumes a lot of time and computing resources, but also easily leads to overfitting to the source domain, thereby reducing its generalization ability. Therefore, the main purpose of us is to study how to fully utilize the original capabilities of CLIP, rather than modifying or fine-tuning it.

Cho et al. [11] synthesize diverse styles to simulate various distribution shifts in the joint feature space, and train a classification head using synthesized style-content features. They utilize learnable style word vectors to achieve diversification of styles in the joint feature space, compensating for the lack of style distribution information during training. Inspired by this, we propose a diverse style prompt assistant method for remote sensing object classification DG (DSPA), as shown in Fig. 1. Although both our method and PromptStyler [11] involve prompt learning, there are significant differences in mechanism and objectives. PromptStyler aims to generate diverse style words, which are then concatenated with class names to obtain text features. They use these features to train a classification head, which is subsequently employed directly for image classification. In contrast, our method leverages diverse text features to guide a lightweight classification model in learning crucial information embedded within these features, thereby acquiring the ability to extract domain-invariant features. Although our approach also utilizes text prompts, the underlying goal is to transfer the richness of text information into the visual feature space by utilizing CLIP's feature alignment properties, thereby reducing the need for image samples when training the classification model.

Our method is divided into two stages. The main task of the first stage is to generate style prompts. We utilize style diversity loss and maximum entropy loss to constrain the learnable style word vectors, generating a series of style prompts representing different styles. The second stage involves classification model training. We combine the style prompts generated in the first stage with class names to obtain complete text prompts, which contain different style and content information. By inputting these text prompts into the CLIP text encoder, text features with different styles and content information can be obtained. Since the text features of CLIP are aligned with the image features in the joint feature space, we are essentially obtaining diverse image features in the joint feature space. We calculate the feature center of each class of text features as the central feature of that class, which is the most general feature of that class without any style bias. Then, we force the image features generated by the classification model to approach the central features of the corresponding category. In this way, we can guide the classification model to generate general image features using only samples from a single domain. Our method utilizes the variability of text to simulate sample diversity, solving the problem of single-sample distribution and improving the generalization performance of the classification model.

### A. Diverse Style Prompt Generation

The goal of the first stage is to use CLIP's text encoder to learn diverse text prompts, and the specific implementation process is shown in Fig. 2. We would like the complete text prompt to be "A [S] style of a [class name].", where "[S]" represents a specific style and "[class name]" represents the class name. In the first stage, we hope to use style diversity loss and maximum entropy loss to learn diverse style prompts, and we hope that the learned style prompts only contain style information and not content information. Therefore, we remove the class name and use "A [S] style of a" as the style prompt and input it to the CLIP text encoder. Among them, "[S]" only exists as a placeholder, and during training, it will be replaced by a learnable style word vector $S_k$. The input of the CLIP text encoder in the first stage is "A $S_k$ style of a," which we denote as $P_S^k, (k = 1, 2, \ldots, K)$. Among them, $S_K$ represents the $k$th style word vector, and K is the number of style prompts to be learned. The style carried by each style prompt $P_S^k$ is different according to the different styles of $S_K$. For the sake of conciseness in expression, we will omit the superscripts when there is no need to specifically distinguish a certain prompt or feature. In order to enable style prompts to learn as much different style information as possible while ignoring content information, and effectively simulate the distribution shifts in the feature space, we design three losses to constrain their training process.

*1) Style Diversity Loss:* Before inputting $P_S$ into the text encoder, we first randomly initialize it. Then, we input the initialized style prompt $P_S$ into the CLIP text encoder to obtain the style feature $F_S$. In order to ensure that the style information learned from each style prompt is different from each other, we add style diversity loss to the style features to constrain the uniqueness of each style feature. We believe that the diversity between different style features is highest when they are orthogonal to each other in the feature space. Therefore, we design the style diversity loss $L_{SD}$ as follows:

$$L_{SD} = \frac{1}{K-1} \sum_{m=2}^{K} \left( \frac{1}{m-1} \sum_{n=1}^{m-1} \left| \frac{F_S^m}{\|F_S^m\|} \cdot \frac{F_S^n}{\|F_S^n\|} \right| \right) \tag{1}$$

where $F_S^m$ and $F_S^n$ are the style features generated by the text encoder after inputting the style prompt $P_S^m$ with the $m$th style word vector $S_m$ and the style prompt $P_S^n$ with the $n$th style word vector $S_n$, respectively, $K$ is the number of style prompts.

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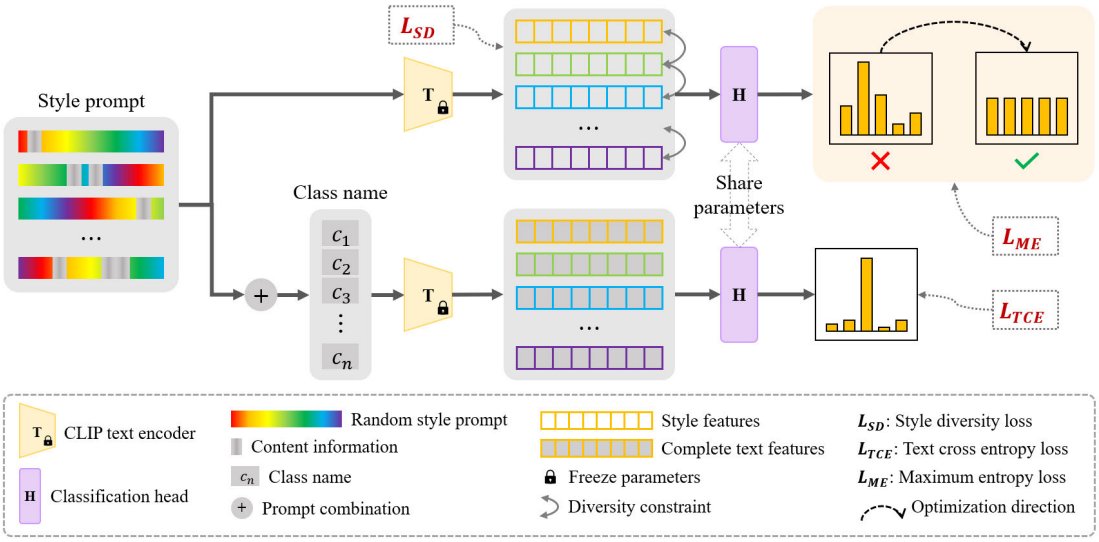



Fig. 2. Algorithm flow of style prompt generation stage. We utilize style diversity loss to constrain the diversity of style prompts, utilize maximum entropy loss to ensure that style prompts do not contain content information, and utilize text cross-entropy loss to avoid learning irrelevant class information.

$L_{SD}$ forces style features to be separated from each other in the feature space until they are orthogonal by minimizing the cosine similarity between every two style features, thereby achieving diversity in style prompts.

*2) Maximum Entropy Loss:* In addition to ensuring diversity between different style prompts, we also need to ensure that each style prompt only contains style information and does not contain any content information related to the category. A very intuitive idea is that if the style prompt $P_S$ only contains style information, then the style feature $F_S$ generated by it should also not contain any information that can distinguish categories. Therefore, if the style feature $F_S$ is input into the classification head to generate the output logits $V_S$, it should also not be able to distinguish any category. Further analysis shows that for the output logits $V_S$, if it cannot distinguish the difference between classes, then the style prompt $P_S$ should exhibit a uniform probability distribution across all classes. In other words, if the style prompt $P_S$ does not contain any content information, then $P_S$ should not belong to any category. According to information theory, the above conditions are equivalent to maximizing the entropy of the predicted distribution; that is, when the entropy of the output logits $V_S$ is maximized, it can meet the requirement that the style prompt does not contain content information. Based on the above analysis, we add a maximum entropy loss $L_{ME}$ to the output logits $V_S$

$$H_{V_S^k} = -\sum_{i=1}^{N} p_S^i log p_S^i \tag{2}$$

$$L_{ME} = -\frac{1}{K}\sum_{k=1}^{K} H_{V_S^k} \tag{3}$$

where $H_{V_S^k}$ represents the entropy of the output logits $V_S^k$ corresponding to the $k$th style prompt $P_S^k$, $p_S^i$ represents the probability that $P_S^k$ belongs to the $i$th class, and $N$ represents the number of classes.

$L_{ME}$ forces each style word vector $S_K$ to only learn style information, so that they are not affected by content information when simulating the diversity of feature space distribution, and can represent a more general style distribution.

*3) Text Cross Entropy Loss:* In addition, in the process of utilizing the aforementioned loss to increase the diversity of style prompts, there may be an extreme situation where the style word vector $S_K$ may learn irrelevant classes orthogonal to the target category. For example, when the target classes are airplane, ship, and car, the style word vector $S_K$ may learn irrelevant classes such as cat and dog. Although this situation also satisfies the constraints of style diversity loss and maximum entropy loss, it does not meet the true learning objectives.

To address the above issues, we propose an additional text cross-entropy loss as a constraint to ensure that the style word vector can truly learn style information rather than irrelevant classes. The specific implementation process is as follows: we combine each style prompt $P_S^k, (k = 1, 2, \ldots, K)$ with the class name $c_n, (n = 1, 2, \ldots, N)$ to obtain $K \times N$ complete text prompts $P_T$ with both style and content information. Then, we input the complete text prompt $P_T$ into the CLIP text encoder to generate text feature $F_T$, which are then processed by the classification head to obtain the output logits $V_T$. Since $P_T$ contains clear content information, the output logits $V_T$ generated by the classification head can correctly distinguish the category. We add a text cross-entropy loss $L_{TCE}$ as a constraint to $V_T$ to ensure that the category can be correctly distinguished when the style prompt is combined with the class name, thus avoiding the above situation. The loss calculation is as follows:

$$L_{TCE} = -\frac{1}{KN}\sum_{i=1}^{KN} l_{CE}\left(V_T^i, Y_T^i\right) \tag{4}$$

where $l_{CE}$ is the cross entropy loss, $V_T^i$ represents the output logits corresponding to the $i$th complete text prompt $P_T^i$, and $Y_T^i$ represents the true probability distribution corresponding

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to $P_T^i$. It is usually represented by a one-hot vector, that is, only the probability corresponding to the true category is equal to 1; the rest are 0. K is the number of styles, and $N$ is the number of categories.

Therefore, the total loss $L_{prompt}$ for the first stage is

$$L_{prompt} = \lambda_{SD} L_{SD} + \lambda_{ME} L_{ME} + \lambda_{TCE} L_{TCE} \tag{5}$$

where $\lambda_{SD}$, $\lambda_{ME}$, and $\lambda_{TCE}$ are the weights corresponding to each loss.

The first stage constrains the optimization process of style prompts through three losses. By using $L_{SD}$, different style information can be learned as much as possible among various style prompts, improving the diversity of style prompts. $L_{ME}$ can make the style prompt only contain style information and not content information. $L_{TCE}$ can prevent style prompts from learning irrelevant classes orthogonal to the target category, while also ensuring that categories can be correctly distinguished when the style prompt is combined with the class name.

In addition, the combination of $L_{SD}$ and $L_{ME}$ also achieves a joint constraint at both the feature level and the logit level. The maximum entropy loss ($L_{ME}$) is usually used to encourage the probability of the model output to tend toward a uniform distribution. In our work, we apply it at the logit level to force the model to assign equal prediction probabilities to each class, thereby preventing the style features from learning content information. Orthogonal loss ($L_{SD}$) is commonly used to enhance the independence and diversity of features. Here, we apply it to style features to encourage them to learn distinct style information. The integration of these two losses establishes a joint constraint on the style prompts, enabling them to learn diverse style information while effectively discarding content information.

By utilizing $L_{prompt}$ as a global constraint, we can obtain diverse style prompts after the first stage, which can serve as important aids in guiding the classification model to learn general representations in the second stage.

### B. Model Optimization

The second stage is the training phase of the classification model, with the main goal of using the learned diverse style prompts to guide the classification model to learn domain-invariant image features. For single-source DG tasks, obtaining diverse image samples is very difficult. However, due to the accessibility and variability of text, we can easily access a variety of texts. If the rich information carried by text can be reasonably utilized to assist in training classification models, it can, to some extent, solve the problem of scarce image samples. Due to the alignment of CLIP's text and image features in the joint feature space, we can generate text features with different style information by introducing diverse style prompts in the joint feature space. By utilizing these diverse text features, we can effectively simulate the distribution changes of image features. In this way, diversity of image data can be achieved in the feature space using only a single source domain, avoiding the need for multiple source domains in traditional methods and reducing the training data requirements for DG tasks.

In order to enable the classification model to fully utilize the knowledge of CLIP and the diversity of text features, we design the following four losses to constrain the training process of the classification model. The training framework is shown in Fig. 3.

*1) Contrast Cross Entropy Loss:* We combine the style prompt $P_S$ obtained in the first stage with the class name $c_n$ to obtain the complete text prompt $P_T$, and then input $P_T$ into the CLIP text encoder to obtain the text feature $F_T$. Different text features can simulate image features with different styles and content information in the joint feature space according to the differences of $P_S$ and $c_n$. We calculate the feature centers of each class of text feature separately as the central feature $F_C$ of the class. Due to the diverse style information and the same content information contained in each class of text feature, the obtained central feature can be regarded as a general feature with the most essential information of the class. Meanwhile, we input the image $X$ of the source domain into the feature encoder E of the classification model to obtain the image features $F_E$. In order to learn the diversity of text features, we compute the cosine similarity between $F_E$ and the central feature $F_C$ of each class to obtain the contrast logits $V_C$. We add a contrast cross-entropy loss $L_{CCE}$ to the contrast logits $V_C$ to reduce the distance between image features and the central features of the corresponding class, thereby promoting the classification model to generate more general image features. The calculation of $L_{CCE}$ is as follows:

$$L_{CCE} = \sum_{i=1}^{B} l_{CE}\left(V_C^i, Y^i\right) \tag{6}$$

where $l_{CE}$ is the cross entropy loss, $V_C^i$ represents the contrast logits corresponding to the $i$th image, $Y^i$ represents the true probability distribution corresponding to the $i$th image, and $B$ is the number of input images for each training batch.

Through the constraint of $L_{CCE}$, the features generated by the classification model can be guided to continuously approach the central features of the corresponding class and move away from the central features of other classes during training.

*2) Image Cross Entropy Loss:* In addition to ensuring that the features generated by the classification model are closer to the general features, it is also necessary to ensure that the classification model has basic classification capabilities. We denote the logits output by the classification head of the classification model as $V_H$, and add an image cross-entropy loss $L_{ICE}$ to it

$$L_{ICE} = \sum_{i=1}^{B} l_{CE}\left(V_H^i, Y^i\right) \tag{7}$$

where $V_H^i$ represents the output logits corresponding to the $i$th image.

*3) Feature Distillation Loss:* In addition, to ensure that the features generated by the classification model can be projected into the same feature space as those of CLIP, we consider the image encoder of CLIP as a teacher and add feature distillation loss $L_{FD}$ to the encoder of the classification model. For the input image $X$, we denote the features generated by

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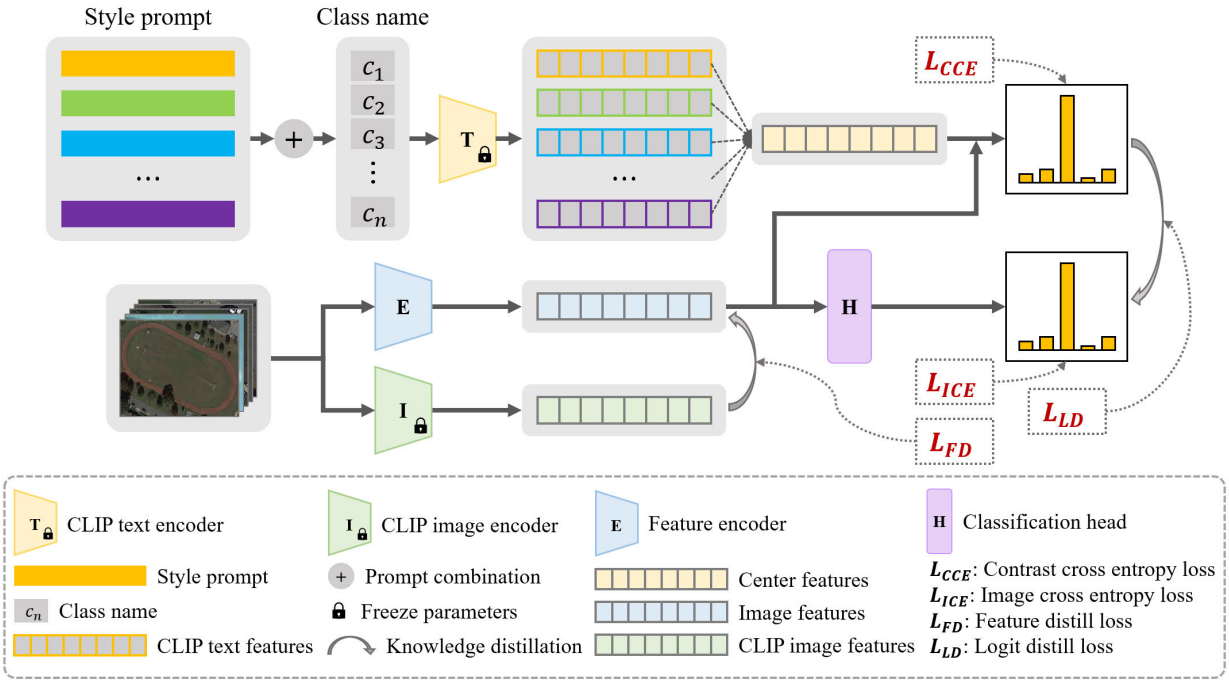



Fig. 3. Algorithm flow of classification model training stage. We utilize contrast cross-entropy loss to force image features to approach the feature centers of corresponding text features, utilize image cross entropy loss to ensure that the classification model has the most basic classification ability, utilize feature distill loss to force the classification model to project image features into the feature space of CLIP, and utilize logits distill loss to constrain the learning process of the classification head.

the CLIP image encoder as $F_I$, and the features generated by the classification model image encoder as $F_E$. Then, we add feature distillation loss between $F_E$ and $F_I$ to force the encoder of the classification model to project features into the joint feature space where CLIP image features and text features are located. The feature distillation loss $L_{FD}$ used is

$$L_{FD} = l_{mse}(F_E, F_I) = \sum_{i=1}^{B} \left|\left|F_E^i - F_I^i\right|\right|_2^2 \tag{8}$$

where $l_{mse}$ is the mean square error loss, $F_E^i$ represents the classification model image features corresponding to the $i$th image, and $F_I^i$ represents the CLIP image features corresponding to the $i$th image.

Due to the pretraining with massive data, the CLIP model has strong feature extraction capabilities. By using $L_{FD}$ as a constraint, it can not only guide the classification model to project features into the joint feature space where the CLIP features are located, but also distill the powerful feature extraction capabilities of the CLIP into the classification model, accelerating the convergence process of training.

It is worth mentioning that we use mse loss ($L_{FD}$) for distillation instead of adding a projection layer. In our method design, the role of distillation is mainly to force the image encoder to project features into the CLIP's feature space, while the core goal of learning domain-invariant features is mainly achieved through the constraint of $L_{CCE}$. Therefore, although there is a significant difference in the architecture between the two distilled networks, directly using mse loss as a strong constraint will not have a negative impact on the learning efficiency of our model. Regarding the strategy of adding a projection layer for alignment, since the CLIP model achieves probability prediction through contrastive learning between text features and image features, there are no preexisting projection layers readily available. Adopting this strategy will require pretraining of the projection layer, thereby introducing additional training steps and computational overhead. In addition, since the $L_{CCE}$ also works at the feature level, applying the distillation loss $L_{FD}$ directly in the latent feature space also promotes greater consistency throughout the learning process, which is more conducive to the alignment between image and text features.

*4) Logit Distillation Loss:* We add feature distillation loss $L_{FD}$ to the intermediate features of the classification model to constrain the learning process of the image encoder in the classification model. Similarly, in order to constrain the learning process of the classification head in the classification model, we add a logit distillation loss $L_{LD}$ to the logits vector $V_H$ output by the classification head. We use the contrast logits $V_C$ obtained by cosine similarity calculation of $F_E$ and $F_C$ as the distillation target, and the logit distillation loss $L_{LD}$ added to $V_H$ is calculated as follows:

$$L_{LD} = l_{KL}(V_H, V_C) \tag{9}$$

$l_{KL}$ is the KL divergence loss.

Therefore, the total loss $L_{model}$ for the second stage is

$$L_{model} = \lambda_{CCE} L_{CCE} + \lambda_{ICE} L_{ICE} + \lambda_{FD} L_{FD} + \lambda_{LD} L_{LD} \tag{10}$$

where $\lambda_{CCE}$, $\lambda_{ICE}$, $\lambda_{FD}$, and $\lambda_{LD}$ are the weights corresponding to each loss.

We use $L_{ICE}$ to assist classification model in learning the basic classification objectives; using $L_{CCE}$ to guide the classification model to generate image features that are close to the

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feature centers of the corresponding text features; using $L_{FD}$ to assist the encoder of the classification model in learning the feature extraction ability of CLIP and forcing it to project the features into the joint feature space of CLIP; utilizing $L_{LD}$ to further constrain the learning process of the classification head. Through the combined effect of these losses, a generalizable classification model capable of generating general features is ultimately obtained.

Our work utilizes a pretrained CLIP model to assist the classification model in learning domain-invariant features using only a single source domain, reducing the model's dependence on diverse image data and achieving DG for remote sensing object classification.

**Algorithm 1** Overall Algorithm Flow

**Input:** Source domain $D_S$, CLIP text encoder $T$, CLIP image encoder $I$, class name $c_n$, total epoch $EP$
**Output:** Optimal classification model
/* Diverse style prompt generation */
Randomly initialize classification head $H$
Randomly initialize style prompt $P_S$
Combine $P_S$ with $c_n$ to obtain text prompt $P_T$
**for** $ep = 1, 2, \ldots, EP$ **do**
  Generate style feature $F_S$ and complete text feature $F_T$, and corresponding logits $V_S$ and $V_T$
  $L_{SD} = \frac{1}{K-1} \sum_{m=2}^{K} \left( \frac{1}{m-1} \sum_{n=1}^{m-1} |\frac{F_S^m}{\|F_S^m\|} \cdot \frac{F_S^n}{\|F_S^n\|}| \right)$
  $L_{ME} = -\frac{1}{K} \sum_{k=1}^{K} H_{V_S^k}$, $H_{V_S^k} = -\sum_{i=1}^{N} p_S^i log p_S^i$
  $L_{TCE} = -\frac{1}{KN} \sum_{i=1}^{KN} l_{CE}(V_T^i, Y_T^i)$
  $L_{prompt} = \lambda_{SD} L_{SD} + \lambda_{ME} L_{ME} + \lambda_{TCE} L_{TCE}$
  Update $P_S$ and $H$ using $L_{prompt}$
**end for**
/* Model optimization */
**for** $ep = 1, 2, \ldots, EP$ **do**
  Combine optimized $P_S$ with $c_n$ to obtain $P_T$
  Generate image feature $F_E$, CLIP text central feature $F_C$, and CLIP image feature $F_I$
  Generate contrast logits $V_C$ by comparing $F_E$ and $F_C$ and classification logits $V_H$ through $H$
  $L_{CCE} = \sum_{i=1}^{B} l_{CE}(V_C^i, Y^i)$
  $L_{ICE} = \sum_{i=1}^{B} l_{CE}(V_H^i, Y^i)$
  $L_{FD} = l_{MSE}(F_E, F_I) = \sum_{i=1}^{B} \|F_E^i - F_I^i\|_2^2$
  $L_{LD} = l_{KL}(V_H, V_C)$
  $L_{model} = \lambda_{CCE} L_{CCE} + \lambda_{ICE} L_{ICE} + \lambda_{FD} L_{FD} + \lambda_{LD} L_{LD}$
  Update feature encoder $E$ and classification head $H$ using $L_{model}$
**end for**
Obtain the optimal Classification model $(E + H)$

The training process is divided into two stages, and the overall flow is shown in Algorithm 1. In the first stage, the optimization process of style prompts is constrained from different aspects through three types of losses, thereby generating diverse style prompts. In the second stage, the optimized style prompts are used as auxiliary information to guide the learning process of the classification model, and the generalization of the classification model is improved through cross-modal feature guidance.

In the testing and inference phase, there is no need to use the CLIP model anymore. The images can be directly input into the trained classification model to obtain the final prediction results.

## IV. Experiment

### A. Experimental Setup

*1) Dataset:* We validate the effectiveness of our work on three commonly used remote sensing image datasets, namely HRRSD [43], DOTA [44], and DIOR [45]. We select the common categories of these three datasets as experimental data. After screening, we retain ten categories as target categories. The final train set samples for HRRSD, DOTA, and DIOR are 10 647, 78 846, and 59 204, while the test set samples are 10 092, 23 621, and 113 899. When conducting ablation experiments, we use HRRSD as the training dataset and DOTA and DIOR as the testing datasets. When validating the effectiveness of our work and comparing it with other methods, we train the model on one dataset and test it on the other two datasets to fully evaluate the performance of the model.

*2) Implementation Detail:* We use PyTorch as the deep learning framework. All of our experiments are conducted using an RTX 2080 Ti GPU. We use Transformer as the text encoder and ViT-B/32 as the image encoder for CLIP. We choose to use a publicly available pretrained CLIP model and freeze the parameters of the text encoder and image encoder throughout the training process. We use ResNet-50 as our classification model and initialize it with its pretrained parameters on ImageNet. It should be noted that in order to ensure that the image features output by the classification model are consistent in dimension with the features output by the CLIP model, we add an FC layer before the last FC layer of ResNet-50 to adjust the feature dimension, while keeping the remaining structure of ResNet-50 unchanged.

The total number of style prompts learned in the first stage is 10, the number of learnable style word vectors is 1, the dimension of each style word vector is set to 512, and the initialization prompt is "A [S] style of a [class name]." In addition, before inputting the style prompts into the text encoder, we randomly initialize the style word vectors using a normal distribution with a mean of 0 and a variance of 0.02. During training and testing, the input images are uniformly adjusted to a size of $224 \times 224$, and the batch sizes are set to 32 and 256, respectively. The number of target classes is 10.

We use AdamW as the optimizer for training style prompts, classification head, and classification model, with learning rates set to 0.001, 0.0001, and 0.0005, respectively, and adjust the learning rate using a cosine annealing scheduler. When using KL divergence loss for distillation learning, we set the temperature to 3.

The loss weights $\lambda_{SD}$, $\lambda_{ME}$, and $\lambda_{TCE}$ are set to 0.5, 0.25, and 0.25, respectively, while $\lambda_{ICE}$, $\lambda_{CCE}$, $\lambda_{FD}$, and $\lambda_{LD}$ are set to 0.25, 0.5, 0.125, and 0.125, respectively. The training epochs for both the style prompt generation stage and the classification model training stage are set to 100. We use Top-1 accuracy to measure the classification performance of the models.

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TABLE I
TOP-1 ACCURACY (%) OF EACH LOSS ABLATION IN THE STYLE PROMPT GENERATION STAGE

| Loss | | | Result | | |
|---|---|---|---|---|---|
| $L_{SD}$ | $L_{ME}$ | $L_{TCE}$ | DOTA | DIOR | Average |
| | ✓ | ✓ | 54.87 | 52.75 | 53.81 |
| ✓ | | ✓ | 53.71 | 52.81 | 53.26 |
| ✓ | ✓ | | 53.71 | 54.83 | 54.27 |
| ✓ | ✓ | ✓ | 57.60 | 53.84 | 55.72 |

TABLE II
TOP-1 ACCURACY (%) OF EACH LOSS ABLATION IN THE CLASSIFICATION MODEL TRAINING STAGE

| Loss | | | | Result | | |
|---|---|---|---|---|---|---|
| $L_{CCE}$ | $L_{ICE}$ | $L_{FD}$ | $L_{LD}$ | DOTA | DIOR | Average |
| | ✓ | ✓ | ✓ | 50.10 | 50.17 | 50.14 |
| ✓ | | ✓ | ✓ | 55.21 | 51.81 | 53.51 |
| ✓ | ✓ | | ✓ | 49.76 | 51.12 | 50.44 |
| ✓ | ✓ | ✓ | | 56.28 | 54.54 | 55.41 |
| ✓ | ✓ | ✓ | ✓ | 57.60 | 53.84 | 55.72 |

### B. Ablation Study

Our work consists of two stages: prompt generation and model training, both of which utilize various losses to constrain the optimization process of prompts or the model. Different losses constrain the parameter update process from different levels. In this section, we validate the effectiveness of each loss through a series of ablation experiments. The ablation experiments conducted in this section are trained on HRRSD and tested on DOTA and DIOR. In addition, in order to better compare the effects of different experimental settings, we also conduct a complete experiment that includes all losses, which is the method we propose. The results are shown in the fourth row of Table I and the fifth row of Table II.

*1) Effectiveness of Style Diversity Loss:* In the style prompt generation stage, in order to increase the diversity of style prompts and ensure that the style information carried by different style prompts is different from each other, we add style diversity loss $L_{SD}$ to the style features output by the text encoder. $L_{SD}$ pushes different style features away from each other in the feature space, forcing different style features to be orthogonal to each other. To verify the effectiveness of the style diversity loss, we remove $L_{SD}$ and only retain the maximum entropy loss and text cross-entropy loss. The results are shown in the first row of Table I. According to the results, after removing the style diversity loss, the Top-1 accuracy of the classification model on DOTA and DIOR decreases by 2.73% and 1.09%, respectively, compared to the complete method. This indicates that the style diversity loss plays an important role in style learning.

*2) Impact of Maximum Entropy Loss:* We use the maximum entropy loss $L_{ME}$ to constrain style prompts to only contain style information and not content information. Adding maximum entropy loss to the output results can ensure that style prompts can simulate more common style distributions in actual samples without bias toward any class. We remove $L_{ME}$ and compare the results with the complete method. The results without $L_{ME}$ are shown in the second row of Table I. By comparison, it can be seen that after removing $L_{ME}$, the Top-1 accuracy of the classification model on DOTA and DIOR decreases to 53.71% and 52.81%, respectively, which indicates the importance of the maximum entropy loss.

*3) Evaluating Text Cross Entropy Loss:* In our previous analysis, we mentioned an extreme situation that needs to be avoided: the style word vectors learn orthogonal classes of the target class rather than style information. To avoid this situation, we design the text cross-entropy loss $L_{TCE}$ as a constraint. Now we remove $L_{TCE}$ to observe whether not constraining this extreme situation will affect the generalization performance of the classification model. The results are shown in the third row of Table I. According to Table I, after removing $L_{TCE}$, the performance of the classification model on DOTA has decreased to a certain extent, and the Top-1 accuracy has decreased by 3.89%. In addition, although the accuracy of the classification model has slightly improved on DIOR, the average accuracy on both datasets has decreased by 1.45% compared to the complete method, indicating that the performance of the model is unstable after removing $L_{TCE}$. This also proves that the occurrence of the extreme situation mentioned above will have a negative impact on the stability and overall performance of the classification model. Although the impact of this extreme situation on model performance is not significant compared to other losses, it does not meet our design goal of learning style information from style word vectors. Therefore, we still chose to add text cross-entropy loss in the complete method we ultimately used to avoid this situation.

*4) Benefits of Contrastive Cross Entropy Loss:* We reduce the distance between image features and corresponding class center features by adding contrastive cross-entropy loss $L_{CCE}$ between the image features generated by the classification model and the center features of the text features generated by CLIP, thereby promoting the classification model to learn general features. $L_{CCE}$ can help classification models learn the stylistic diversity of text features, which is of great significance for learning domain-invariant features. To verify the importance of $L_{CCE}$ for general feature learning, we remove $L_{CCE}$ to observe the changes in the generalization performance of the classification model. The results are shown in the first row of Table II. Through the results, it can be found that after removing $L_{CCE}$, the generalization performance of the classification model decreases, and the Top-1 accuracy on DOTA and DIOR decreases to 50.10% and 50.17%, respectively. This demonstrates the importance of contrastive cross-entropy loss $L_{CCE}$ in domain-invariant feature learning.

*5) Efficacy of Image Cross Entropy Loss:* In order to ensure that the classification model has the basic ability to solve the target task, we add an image cross-entropy loss $L_{ICE}$ to the output of the classification model to constrain its basic learning direction. Similarly, we also conduct ablation experiments on $L_{ICE}$, and the results are shown in the second row of Table II. By comparing the results with the complete method,

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TABLE III
TOP-1 ACCURACY (%) CORRESPONDING TO DIFFERENT PROMPT TEMPLATES. FOR BREVITY, [CLASS NAME] IS ABBREVIATED AS [CLS] IN THE TABLE

| prompt template | Result | | |
|---|---|---|---|
| | DOTA | DIOR | Average |
| "[S]-style [cls]." | 53.44 | 51.98 | 52.71 |
| "A [S] [cls] photo." | 53.28 | 51.91 | 52.60 |
| "A [S] style of a [cls]." | **57.60** | 53.84 | **55.72** |
| "A [S1] [S2] style of a [cls]." | 55.29 | **54.62** | 54.96 |
| "A [S1]...[S4] style of a [cls]." | 55.99 | 52.09 | 54.04 |
| "A [S1]...[S8] style of a [cls]." | 49.49 | 49.77 | 49.63 |

it can be seen that removing the image cross-entropy loss will affect the performance of the classification model. The Top-1 accuracy on DOTA and DIOR decreases by 2.39% and 2.03%, respectively, compared to the results with $L_{ICE}$ retained. Therefore, the basic image cross-entropy loss $L_{ICE}$ is essential.

*6) Role of Feature Distillation Loss:* In feature learning, in order to force the features generated by the classification model to be projected onto the feature space of CLIP, we add feature distillation loss $L_{FD}$ to the image features generated by the classification model. $L_{FD}$ considers the image encoder of CLIP as a teacher and the encoder of the classification model as a student to constrain the learning process of the classification model encoder. We study the performance of the classification model without feature distillation loss, and the results are shown in the third row of Table II. The Top-1 accuracy of the classification model on DOTA and DIOR is 49.76% and 51.12%, respectively. From the results, it can be seen that the lack of feature distillation loss will reduce the accuracy of the classification model in unseen domains.

*7) Logit Distillation Loss Performance:* Similarly, the purpose of adding the logit distillation loss $L_{LD}$ is to constrain the learning process of the classification head. According to the fourth row of Table II, after removing the logit distillation loss, the average Top-1 accuracy of the classification model on DOTA and DIOR decreases compared to the complete method, with values of 55.41%. Therefore, the effectiveness of logit distillation loss can be demonstrated.

### C. Sensitivity Analysis

*1) Prompt Template:* During the style prompt generation stage, we utilize text sentences without class names as the initialized style prompt template. We replace the placeholder in the style template with learnable style word vectors to obtain style prompts, and concatenate them with class names to obtain complete text prompts. To evaluate the model's sensitivity to the choice of prompt template, we conduct ablation studies using three different templates, and the results are shown in the upper part of Table III. From the results, it can be seen that the classification model maintains robust accuracy on the test datasets regardless of the template employed, with the template "A [S] style of a [class name]." yielding the best performance.

TABLE IV
TOP-1 ACCURACY (%) UNDER DIFFERENT VALUES OF $K$

| prompt num | Result | | |
|---|---|---|---|
| | DOTA | DIOR | Average |
| 4 | 57.19 | 53.18 | 55.19 |
| 6 | 56.89 | 51.68 | 54.29 |
| 8 | 53.45 | 53.44 | 53.45 |
| 10 | **57.60** | **53.84** | **55.72** |
| 12 | 56.82 | 53.43 | 55.13 |

In addition, the number of placeholders within the template is also worth studying. We use "A [S] style of a [class name]." as the template and vary the count of placeholders "[S]" in the template to observe the corresponding performance. The results are shown in the lower part of Table III. A comparison of the results on the test datasets reveals that model performance exhibits some fluctuation with different numbers of placeholders, and gradually declines as the number increases. This observation aligns with our theoretical rationale. Since our goal is to ensure that each style prompt only learns a single, distinct style to prevent knowledge entanglement. As the number of placeholders increases, the style information learned by each style prompt may become less pure, thereby weakening the final learning effect.

*2) Number of Style Prompts:* In the style prompt generation stage, we train $K$ style prompts to simulate $K$ different styles. To investigate the relationship between model performance and the setting of $K$, we train models with different values of $K$ and record the corresponding results, as shown in Table IV. By comparing the results under different $K$ values, it can be seen that when the value of $K$ changes, the model performance is relatively stable, that is, the model is not sensitive to the value of $K$. And when $K$ is set to 10, the model achieves the best performance.

*3) Weight Coefficient of Losses:* At different stages of learning, we apply different loss functions to constrain the training process in various aspects. The style prompt generation stage requires constraints on prompt diversity, decoupling from content information, and semantic plausibility. Therefore, $L_{SD}$, $L_{ME}$ and $L_{TCE}$ are designed, respectively. During the classification model optimization stage, it is important to ensure that the image encoder learns the information from text features; thus, $L_{CCE}$ and $L_{FD}$ are added. At the same time, it is also necessary to constrain the learning process of the classification head, so we design $L_{ICE}$ and $L_{LD}$. While all these losses play crucial but distinct roles, their relative importance varies. Therefore, we assign an independent weight coefficient to each loss to maximize its individual contributions.

When determining the relative importance of these loss functions, in order to avoid the training process relying too much on manual hyperparameter tuning and weakening the practicality of the model, we use a combination of theoretical analysis and parameter search to determine the values of each weight coefficient. Specifically, we first make a preliminary assessment of the importance of each loss function based

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TABLE V
TOP-1 ACCURACY (%) UNDER DIFFERENT VALUES OF $\lambda_{SD}$

| $\lambda_{SD}$ | Result | | |
|---|---|---|---|
| | DOTA | DIOR | Average |
| 0.25 | 57.03 | **54.30** | 55.67 |
| 0.5 | **57.60** | 53.84 | **55.72** |
| 0.75 | 54.79 | 54.28 | 54.54 |
| 1 | 52.79 | 52.85 | 52.82 |

TABLE VI
TOP-1 ACCURACY (%) UNDER DIFFERENT VALUES OF $\lambda_{CCE}$

| $\lambda_{CCE}$ | Result | | |
|---|---|---|---|
| | DOTA | DIOR | Average |
| 0.25 | 55.85 | 54.06 | 54.96 |
| 0.5 | **57.60** | 53.84 | **55.72** |
| 0.75 | 54.18 | **54.76** | 54.47 |
| 1 | 52.18 | 52.05 | 52.12 |

on our design objectives and the desired model optimization direction. In the style generation stage, the highest priority is to ensure that the learned style prompts are diverse to avoid model collapse. Therefore, $L_{SD}$ is assigned a larger weight. The subsequent constraints $L_{ME}$ and $L_{TCE}$ used to improve learning quality are assigned relatively smaller weights. In the model optimization stage, the primary goal is to ensure that image features adequately learn from text features. Hence, $L_{CCE}$ is assigned a relatively large weight. Meanwhile, $L_{ICE}$ is mainly used to constrain the entire classification model to ensure its fundamental learning objective is correct, so it also requires a relatively large weight. Finally, the coefficients of $L_{FD}$ used for auxiliary feature learning and $L_{LD}$ used for auxiliary classification head learning can be set to slightly smaller values. Following this initial estimation, we conduct a series of parameter searches and result analysis to finalize the specific weight coefficients for each loss function.

Ablation studies on two of the key weights, $\lambda_{SD}$ (the weight for $L_{SD}$) and $\lambda_{CCE}$ (the weight for $L_{CCE}$), are presented in Tables V and VI, respectively. As shown in Table V, the performance of the model is relatively stable when $\lambda_{SD}$ is below 0.5, but gradually declines as it increases further. This is because an excessively high weight for $\lambda_{SD}$ diminishes the constraining effect of the other losses. Similarly, the model's accuracy first increases and then decreases as the weight coefficient $\lambda_{CCE}$ for $L_{CCE}$ rises, for the same reason explained above.

As shown in Tables V and VI, the performance of the model remains relatively stable across different weight coefficients. While some fluctuation occurs with weight changes, the model's performance generally remains in a relatively stable state. Therefore, it can be concluded that the model's performance is stable under different hyperparameter combinations and does not heavily rely on manual parameter tuning. Furthermore, the seven losses we proposed are primarily designed to constrain the optimization direction of the model and do not involve data balancing or regularization. The weight coefficients for these losses mainly affect the relative strength of each loss function, and are not tailored to any specific dataset. Instead, they aim to help the model adapt to distribution differences across datasets. Therefore, there is no need to readjust the values of these hyperparameters when applying the model to different datasets or tasks. Our model's excellent performance on three different datasets, HRRSD, DOTA, and DIOR, also confirms this.

### D. Visualization

As previously stated, style-diversified text features can help our classification model learn domain-invariant features. To compare the ability of different methods to learn domain-invariant knowledge, we conduct visualizations for our method and the comparative methods. Taking training on DOTA and testing on DIOR as an example, we extract image features from different models. We utilize t-distributed stochastic neighbor embedding (t-SNE) to visualize the distribution of image features, as illustrated in Fig. 4. Through comparison, it can be seen that our method maintains the optimal interclass separability and intraclass compactness on the test dataset. This indicates that our method has better generalization ability compared to other methods when facing domain shift. Specifically, such intraclass aggregation suggests that our model can extract the most general information for each class to generate domain-invariant features when applied to unseen domains. Meanwhile, the clear boundaries between different classes indicate that our model has a strong discriminative capability for different categories. Therefore, the visualization results of Fig. 4 provide further evidence for the effectiveness of our method in domain-invariant feature learning.

### E. Comparison With State of the Arts

To verify the effectiveness of our work, we compare it with state-of-the-art methods. We select nine different methods as comparison methods, including PDEN [46], ACVC [47], ITTA [48], RIDG [49], MIRO [50], WCD [51], CFCL [52], DPL [24], and APPLeNet [53]. PDEN designs a progressive domain extension network to improve the model's generalization ability. ACVC simulates a new domain by changing the image and uses an attention consistency loss to force class activation map alignment between the original image and its corrupted versions. ITTA proposes an improved test-time adaptive method that utilizes a learnable consistency loss and adaptive parameters updated only during testing to enhance the DG ability of the model. RIDG introduces a rationale concept, which forces the rationale matrix of the sample to be consistent with its corresponding mean value, and uses a rationale invariance loss to achieve model generalization. MIRO solves the DG problem by introducing mutual information with the oracle model. They use a pretrained model to approximate the oracle model to derive a tractable variational bound of the objective. WCD is a weakly correlated distillation learning framework that selects weakly correlated features from teacher models and distills them into the student model. CFCL proposes a center-wise feature consistency learning method, in which the student

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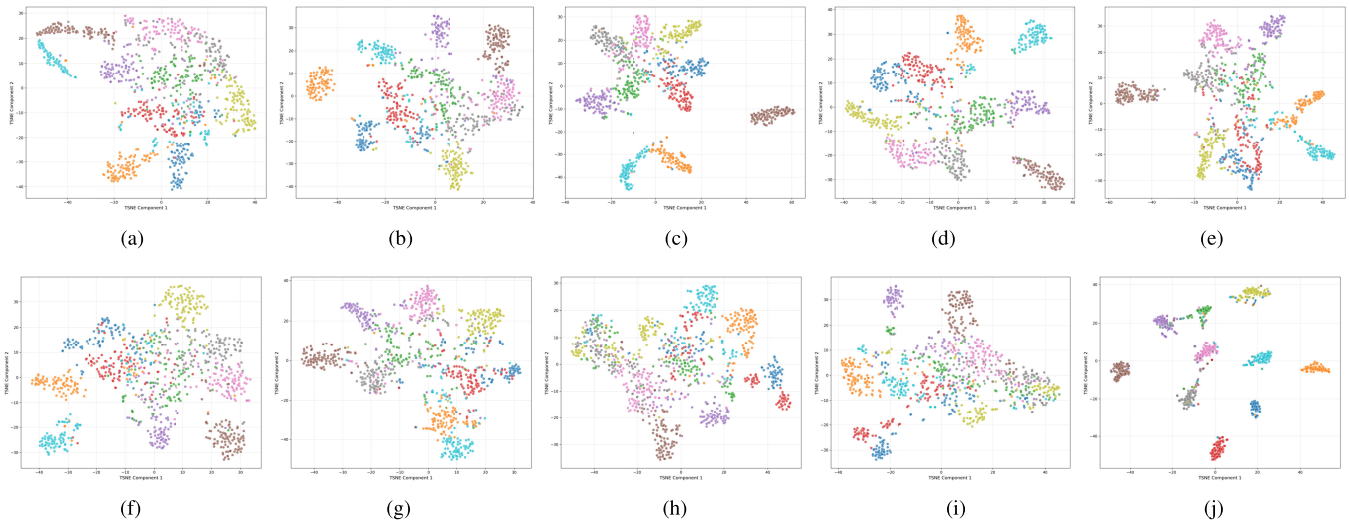



Fig. 4. Visualization of feature embedding from different methods. We use DOTA as the training dataset and DIOR as the test dataset, randomly selecting 100 test samples from each class for visualization. Points of different colors represent samples from different classes. (a) PDEN. (b) ACVC. (c) ITTA. (d) RIDG. (e) MIRO. (f) WCD. (g) CFCL. (h) DPL. (i) APPLeNet. (j) Ours.

TABLE VII

TOP-1 ACCURACY (%) FOR THE MODELS TRAINED ON HRRSD AND TESTED ON DOTA AND DIOR. THE BEST AND SECOND BEST RESULTS ARE MARKED BY BOLDING AND UNDERLINING, RESPECTIVELY

| method | DOTA | DIOR | Average |
|---|---|---|---|
| PDEN | 45.28 | 45.90 | 45.59 |
| ACVC | 47.78 | 44.62 | 46.20 |
| ITTA | 47.01 | <u>53.16</u> | 50.09 |
| RIDG | <u>49.76</u> | 51.94 | <u>50.85</u> |
| MIRO | 48.19 | 49.42 | 48.81 |
| WCD | 45.20 | 51.35 | 48.28 |
| CFCL | 38.00 | 48.35 | 43.18 |
| DPL | 44.59 | 49.60 | 47.10 |
| APPLeNet | 36.63 | 41.30 | 38.97 |
| Ours | **57.60** | **53.84** | **55.72** |

TABLE VIII

TOP-1 ACCURACY (%) FOR THE MODELS TRAINED ON DOTA AND TESTED ON HRRSD AND DIOR. THE BEST AND SECOND BEST RESULTS ARE MARKED BY BOLDING AND UNDERLINING, RESPECTIVELY

| method | HRRSD | DIOR | Average |
|---|---|---|---|
| PDEN | 63.71 | 82.56 | 73.14 |
| ACVC | 71.84 | 69.69 | 70.77 |
| ITTA | <u>76.29</u> | <u>86.01</u> | <u>81.15</u> |
| RIDG | 64.28 | 81.15 | 72.72 |
| MIRO | 61.29 | 81.30 | 71.30 |
| WCD | 46.34 | 75.76 | 61.05 |
| CFCL | 57.96 | 83.84 | 70.90 |
| DPL | 64.0 | 77.82 | 70.93 |
| APPLeNet | 49.03 | 52.90 | 50.97 |
| Ours | **81.42** | **89.72** | **85.57** |

model estimates the central features by referencing those correctly classified by the head and tail teacher models. DPL utilizes a diverse text-prompt generation learning method to suppress the noise generated by inadequate samples in remote sensing image classification tasks. APPLeNet proposes an image-conditioned prompt learning strategy that disentangles visual style and content primitives by introducing an attention-driven injection module and an anti-correlation regularizer.

We train models on one dataset and then test them on the other two datasets to calculate the classification accuracy of different models in the face of unseen domains, thereby comparing their generalization performance.

*1) Training on HRRSD:* We train the models of each comparison method on the training set of HRRSD, and then test them on the test sets of DOTA and DIOR. The test results are shown in Table VII. From the results, it can be seen that our work has higher Top-1 accuracy than the other nine methods on DOTA and DIOR, and the average accuracy on the two datasets has been improved by 10.13%, 9.52%, 5.63%, 4.87%, 6.91%, 7.44%, 12.54%, 8.62%, and 16.75%, respectively. The average accuracy of the second-best method is 50.85%, which is 4.87% lower than our method.

*2) Training on DOTA:* We train the models on the DOTA training set and then test them on the HRRSD and DIOR test sets. The results are shown in Table VIII. According to the results, the average classification accuracy of our method on HRRSD and DIOR is 85.57%, achieving the best results. Especially, our method achieves accuracies of 81.42% and 89.72% on HRRSD and DIOR, respectively, which are much higher than the corresponding accuracies of 76.29% and 86.01% for the suboptimal method ITTA.

*3) Training on DIOR:* The results of training the models on the DIOR training set and testing them on the HRRSD and DOTA test sets are shown in Table IX. According to the results in the table, our work demonstrates the highest Top-1 accuracy on both datasets, with results of 84.01% and 87.89%,

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TABLE IX

TOP-1 ACCURACY (%) FOR THE MODELS TRAINED ON DIOR AND TESTED ON HRRSD AND DOTA. THE BEST AND SECOND BEST RESULTS ARE MARKED BY BOLDING AND UNDERLINING, RESPECTIVELY

| method | HRRSD | DOTA | Average |
|---|---|---|---|
| PDEN | 56.32 | 82.67 | 69.50 |
| ACVC | 74.94 | 78.02 | 76.48 |
| ITTA | <u>75.78</u> | <u>85.16</u> | <u>80.47</u> |
| RIDG | 69.15 | 80.53 | 74.84 |
| MIRO | 67.14 | 81.63 | 74.39 |
| WCD | 70.11 | 83.22 | 76.67 |
| CFCL | 61.13 | 81.33 | 71.23 |
| DPL | 74.61 | 75.49 | 75.05 |
| APPLeNet | 58.26 | 54.29 | 56.28 |
| Ours | **84.01** | **87.89** | **85.95** |

respectively. In addition, the average accuracy of our method reaches 85.95% on two datasets, which is 5.48% higher than the second-best method.

The results obtained from training on three datasets fully demonstrate the effectiveness of our method, which can indeed achieve generalization of the classification model and greatly improve its accuracy in unseen domains. In addition, to further validate our method, we also plot confusion matrices corresponding to each method. Please refer to the supporting materials for the results and analysis of the confusion matrices.

## V. CONCLUSION

We propose a cross-modal prompt-guided feature learning strategy for solving the DG problem in remote sensing object classification. Our proposed strategy is divided into two stages. The first stage simulates the style diversity within the joint feature space of CLIP by generating diverse style prompts. In the second stage, the generated style prompts are used to guide the optimization direction of image features, thereby promoting the classification model to learn domain-invariant features. Our method utilizes textual information to constrain the image processing process, achieving single-source DG for remote sensing object classification. In addition, we conduct extensive experiments on three datasets to demonstrate the effectiveness of our method. Furthermore, the algorithm we proposed mainly leverages the alignment nature of image and text features within the vision-language model to facilitate learning, without introducing any modifications or additional constraints to the visual-language model itself. And this feature alignment property is fundamental and shared across various visual-language models. Consequently, our method is robust to the choice of the underlying visual-language model; that is, our method is readily applicable to other vision-language models as well. We welcome and encourage interested readers to explore the generalization and extension of our method.

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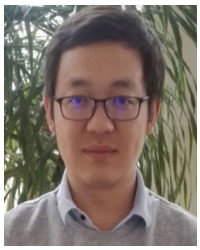

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**Diedie Jia** received the B.S. and M.S. degrees from Dalian University of Technology, Dalian, China, in 2022 and 2025, respectively.

Her research interests include object classification and domain generalization.



**Wenda Zhao** (Member, IEEE) received the B.S. degree from Jilin University, Changchun, China, in 2011, and the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently a Professor with the School of Information and Communication Engineering, Dalian University of Technology, Dalian, China. His research interests include deep learning and computer vision tasks.



**Hao Liu** is currently pursuing the Ph.D. degree with Shanghai Jiao Tong University, Shanghai, China.

He is a Researcher with Wuhan Digital Engineering Research Institute, Wuhan, China. His research focuses on information fusion and situation cognition.



**Haipeng Wang** received the Ph.D. degree from Naval Aviation University, Yantai, China, in 2012.

He is currently a Professor with the Unit 92728 of PLA, Shanghai, China. His research interests include the general area of intelligent perception and fusion, and big data technology and applications.

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